Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Tony Lopes Dairy LP

Physical address of dairy:

Number and Street	City	County	Zip Code
27500 W Bunker RD	Gustine	Merced	95322

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1972

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X070-X010-X012-XXXX X070-X010-X013-XXXX

B. OPERATORS

Lopes, Paul

Operator name: Lopes, Paul

Telephone no.: (209) 649-4967 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
9270 S Whitworth RD	Gustine	CA	95322

This operator is responsible for paying permit fees.

C. OWNERS

Lopes, Paul

Legal owner name: Lopes, Paul

Telephone no.: (209) 649-4967 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
9270 S Whitworth RD	Gustine	CA	95322

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	350	900	900	300	0
Number under roof	2,175	0	0	0	0	300
Maximum number	2,400	350	1,000	1,000	360	360
Average number	2,175	350	950	950	310	310
Avg live weight (lbs)	1,400	1,450	900	675		

Predominant milk cow breed: Holstein

Average milk production: 72 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 82,104.62 *tons per reporting period*

Total nitrogen from manure: 1,011,993.62 *lbs per reporting period*

After ammonia losses (30% loss applied): 708,395.53 *lbs per reporting period*

Total phosphorus from manure: 165,904.99 *lbs per reporting period*

Total potassium from manure: 419,390.52 *lbs per reporting period*

Total salt from manure: 1,104,581.25 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 51,840,000 *gallons*

Total nitrogen generated: 189,470.89 *lbs*

Total phosphorus generated: 26,696.92 *lbs*

Total potassium generated: 310,242.73 *lbs*

Total salt generated: 2,726,772.14 *lbs*

	51,840,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	51,840,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Calf Barn	Ground water
CCID	Ground water
IW 12	Ground water
IW 2	Ground water
IW 3	Ground water

Source Description	Type
IW 7	Ground water
IW 8	Ground water
IW 9	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
06/08/2023	Solid commercial fertilizer *21-0-0*	58.35 *ton*	Dry-weight	0.1	0.210000	0.000000	0.000000	0.000000
06/25/2023	Solid commercial fertilizer *UN 32*	27.80 *ton*	Dry-weight	0.1	0.320000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	422.57	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	422.57	0.00	0.00	0.00

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	Separator solids	880.00 *ton*	Dry-weight	59.1		13,900.00	3,400.00	21,600.00		64.70
05/17/2023	Separator solids	1,300.00 *ton*	As-is	59.1		13,900.00	3,400.00	21,600.00		64.70
10/19/2023	Corral solids	270.00 *ton*	Dry-weight	58.7		20,100.00	3,050.00	4,840.00		67.70
10/19/2023	Corral solids	820.00 *ton*	Dry-weight	58.7		20,100.00	3,050.00	4,840.00		67.70
10/19/2023	Corral solids	430.00 *ton*	Dry-weight	58.7		20,100.00	3,050.00	4,840.00		67.70

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	71,381.73	15,116.79	77,785.26	2,003,743.32

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	71,381.73	15,116.79	77,785.26	2,003,743.32

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 10	100	100	2	process wastewater	X070-X010-X013-XXXX
Field 11	83	83	1	process wastewater	X070-X040-X015-XXXX
Field 12	72	72	2	both	X070-X040-X023-XXXX
Field 14	117	117	2	both	X070-X040-X024-XXXX X070-X040-X025-XXXX
Field 15	38	38	0	none	X070-X060-X023-XXXX
Field 16	138	138	0	none	X063-X250-X022-XXXX
Field 17	139	139	2	manure	X070-X020-X029-XXXX X070-X020-X033-XXXX X070-X020-X044-XXXX
Field 2	86	86	2	process wastewater	X070-X001-X006-XXXX
Field 6	83	83	2	process wastewater	X070-X040-X016-XXXX
Field 7	94	94	2	process wastewater	X070-X010-X005-XXXX
Field 8	26	26	2	process wastewater	X070-X010-X012-XXXX
Field 9	60	60	2	both	X070-X010-X013-XXXX
Totals for areas that were used for application	860	860	19		
Totals for areas that were not used for application	176	176	0		
Land application area totals	1,036	1,036	19		

B. CROPS AND HARVESTS

Field 10

Field name: Field 10

10/27/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 100 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/17/2023	1,254.00 *ton*	As-is		70.0	6,160.00	780.00	7,400.00		8.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	12.54	154.49	19.56	185.59	647.06

Field 10

05/19/2023: Corn, silage

Crop: Corn, silage Acres planted: 100 Plant date: 05/19/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/10/2023	1,885.00 *ton*	As-is		66.6	4,040.00	710.00	4,860.00		10.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	18.85	152.31	26.77	183.22	1,271.77

Field 11

Field name: Field 11

01/01/2023: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 83 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	311.00 *ton*	As-is		10.0	30,900.00	2,550.00	28,400.00		11.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	7.00	420.00	37.80	294.00	0.00
Total actual harvest content	3.75	231.56	19.11	212.83	755.39

Field 12

Field name: Field 12

Field 12

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 72 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/20/2023	338.00 *ton*	As-is		70.0	15,300.00	3,120.00	30,600.00		9.44

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	4.69	143.65	29.29	287.30	265.89

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 72 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	1,587.00 *ton*	As-is		66.6	4,210.00	670.00	4,450.00		3.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	22.04	185.59	29.54	196.17	544.78

Field 14

Field name: Field 14

11/04/2022: Oats, hay

Crop: Oats, hay Acres planted: 117 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/22/2023	518.00 *ton*	As-is		70.0	12,500.00	2,700.00	27,300.00		9.48

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	160.00	26.00	132.00	0.00
Total actual harvest content	4.43	110.68	23.91	241.73	251.83

Field 14

06/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 117 Plant date: 06/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/15/2023	2,164.00 *ton*	As-is		69.6	3,740.00	570.00	3,800.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	18.50	138.35	21.09	140.57	742.20

Field 17

Field name: Field 17

11/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 139 Plant date: 11/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/18/2023	1,294.00 *ton*	As-is		70.0	3,810.00	1,230.00	15,500.00		9.95

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	9.31	70.94	22.90	288.59	555.77

05/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 139 Plant date: 05/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/08/2023	3,473.00 *ton*	As-is		67.0	3,960.00	810.00	5,180.00		7.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	24.99	197.89	40.48	258.85	1,170.83

Field 2

Field name: Field 2

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 86 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/18/2023	1,482.00 *ton*	As-is		70.0	3,960.00	780.00	8,100.00		10.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	17.23	136.48	26.88	279.17	1,044.29

05/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 86 Plant date: 05/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/01/2023	2,226.00 *ton*	As-is		65.1	4,570.00	850.00	4,460.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	25.88	236.58	44.00	230.88	1,156.28

Field 6

Field name: Field 6

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 83 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/16/2023	852.00 *ton*	As-is		70.0	5,400.00	1,350.00	17,600.00		13.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	10.27	110.86	27.72	361.33	849.95

Field 6

04/28/2023: Corn, silage

Crop: Corn, silage Acres planted: 83 Plant date: 04/28/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/26/2023	1,438.00 *ton*	As-is		52.3	5,440.00	980.00	6,270.00		6.69

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	17.33	188.50	33.96	217.26	1,105.75

Field 7

Field name: Field 7

11/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 94 Plant date: 11/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/16/2023	1,394.00 *ton*	As-is		70.0	5,520.00	1,290.00	22,000.00		14.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	14.83	163.72	38.26	652.51	1,254.60

05/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 94 Plant date: 05/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/04/2023	2,496.00 *ton*	As-is		63.7	3,740.00	660.00	3,670.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	26.55	198.62	35.05	194.90	1,079.55

Field 8

Field name: Field 8

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 26 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/15/2023	321.00 *ton*	As-is		70.0	4,320.00	690.00	7,000.00		5.47

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	12.35	106.67	17.04	172.85	405.20

05/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 26 Plant date: 05/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/30/2023	717.00 *ton*	As-is		63.5	4,710.00	940.00	4,610.00		6.39

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	27.58	259.77	51.84	254.26	1,286.38

Field 9

Field name: Field 9

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 60 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/17/2023	1,332.00 *ton*	As-is		70.0	5,540.00	840.00	7,930.00		8.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	150.00	24.00	124.50	0.00
Total actual harvest content	22.20	245.98	37.30	352.09	1,105.56

Field 9

06/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 60 Plant date: 06/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	1,879.00 *ton*	As-is		70.2	3,750.00	690.00	5,660.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	24.00	192.00	36.00	158.40	0.00
Total actual harvest content	31.32	234.88	43.22	354.50	1,175.88

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 10 - 10/27/2022: Oats, silage-soft dough

Field name: Field 10

Crop: Oats, silage-soft dough — Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/14/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	87.48	13.94	150.93	1,211.29	2,880,000.00 *gal*
IW 9	Ground water	6.84	0.00	0.00	490.29	8,640,000.00 *gal*
Application event totals		94.32	13.94	150.93	1,701.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	80.75	12.40	124.73	1,023.83	2,160,000.00 *gal*
IW 9	Ground water	5.13	0.00	0.00	367.71	6,480,000.00 *gal*
Application event totals		85.88	12.40	124.73	1,391.55	

Field 10 - 05/19/2023: Corn, silage

Field name: Field 10

Crop: Corn, silage — Plant date: 05/19/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 9	Ground water	10.41	0.00	0.00	745.64	13,140,000.00 *gal*
Application event totals		10.41	0.00	0.00	745.64	

Field 10 - 05/19/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 9	Ground water	11.97	0.00	0.00	858.00	15,120,000.00 *gal*
Application event totals		11.97	0.00	0.00	858.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	94.21	14.47	145.52	1,194.47	2,520,000.00 *gal*
IW 9	Ground water	11.97	0.00	0.00	858.00	15,120,000.00 *gal*
Application event totals		106.19	14.47	145.52	2,052.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 9	Ground water	8.55	0.00	0.00	612.86	10,800,000.00 *gal*
Application event totals		8.55	0.00	0.00	612.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 9	Ground water	10.26	0.00	0.00	735.43	12,960,000.00 *gal*
Application event totals		10.26	0.00	0.00	735.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 9	Ground water	8.55	0.00	0.00	612.86	10,800,000.00 *gal*
Application event totals		8.55	0.00	0.00	612.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 9	Ground water	10.26	0.00	0.00	735.43	12,960,000.00 *gal*
Application event totals		10.26	0.00	0.00	735.43	

Field 11 - 01/01/2023: Alfalfa, hay

Field name: Field 11

Crop: Alfalfa, hay

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	194.59	29.88	300.56	2,467.06	4,320,000.00 *gal*
IW 9	Ground water	6.18	0.00	0.00	443.03	6,480,000.00 *gal*
Application event totals		200.77	29.88	300.56	2,910.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 9	Ground water	12.37	0.00	0.00	886.06	12,960,000.00 *gal*
Application event totals		12.37	0.00	0.00	886.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 9	Ground water	9.27	0.00	0.00	664.54	9,720,000.00 *gal*
Application event totals		9.27	0.00	0.00	664.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 9	Ground water	11.34	0.00	0.00	812.22	11,880,000.00 *gal*
Application event totals		11.34	0.00	0.00	812.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 9	Ground water	12.37	0.00	0.00	886.06	12,960,000.00 *gal*
Application event totals		12.37	0.00	0.00	886.06	

Field 11 - 01/01/2023: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 9	Ground water	10.30	0.00	0.00	738.38	10,800,000.00 *gal*
Application event totals		10.30	0.00	0.00	738.38	

08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 9	Ground water	10.30	0.00	0.00	738.38	10,800,000.00 *gal*
Application event totals		10.30	0.00	0.00	738.38	

Field 12 - 11/03/2022: Oats, silage-soft dough

Field name: Field 12

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	174.47	26.79	269.49	2,211.98	3,360,000.00 *gal*
IW 12	Ground water	3.31	0.00	0.00	304.93	6,048,000.00 *gal*
Application event totals		177.77	26.79	269.49	2,516.91	

04/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 12	Ground water	4.61	0.00	0.00	424.72	8,424,000.00 *gal*
Application event totals		4.61	0.00	0.00	424.72	

11/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 12	Ground water	4.25	0.00	0.00	392.05	7,776,000.00 *gal*
Application event totals		4.25	0.00	0.00	392.05	

Field 12 - 06/08/2023: Corn, silage

Field name: Field 12

Crop: Corn, silage

Plant date: 06/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 12	Ground water	5.67	0.00	0.00	522.73	10,368,000.00 *gal*
Application event totals		5.67	0.00	0.00	522.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	113.70	27.81	176.69	5,292.46	720.00 *ton*
Application event totals		113.70	27.81	176.69	5,292.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
21-0-0	Solid commercial fertilizer	52.50	0.00	0.00	0.00	
Application event totals		52.50	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 12	Ground water	5.67	0.00	0.00	522.73	10,368,000.00 *gal*
Application event totals		5.67	0.00	0.00	522.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 12	Ground water	4.25	0.00	0.00	392.05	7,776,000.00 *gal*
Application event totals		4.25	0.00	0.00	392.05	

Field 12 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 12	Ground water	4.96	0.00	0.00	457.39	9,072,000.00 *gal*
Application event totals		4.96	0.00	0.00	457.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 12	Ground water	4.25	0.00	0.00	392.05	7,776,000.00 *gal*
Application event totals		4.25	0.00	0.00	392.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 12	Ground water	4.84	0.00	0.00	446.50	8,856,000.00 *gal*
Application event totals		4.84	0.00	0.00	446.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 12	Ground water	4.25	0.00	0.00	392.05	7,776,000.00 *gal*
Application event totals		4.25	0.00	0.00	392.05	

Field 14 - 11/04/2022: Oats, hay

Field name: Field 14

Crop: Oats, hay

Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	138.04	21.20	213.22	1,750.14	4,320,000.00 *gal*
IW 12	Ground water	3.05	0.00	0.00	281.47	9,072,000.00 *gal*
Application event totals		141.09	21.20	213.22	2,031.61	

Field 14 - 11/04/2022: Oats, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 12	Ground water	4.36	0.00	0.00	402.10	12,960,000.00 *gal*
Application event totals		4.36	0.00	0.00	402.10	

12/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 12	Ground water	3.93	0.00	0.00	361.89	11,664,000.00 *gal*
Application event totals		3.93	0.00	0.00	361.89	

Field 14 - 06/24/2023: Corn, silage

Field name: Field 14

Crop: Corn, silage

Plant date: 06/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	116.62	28.53	181.22	5,428.16	1,200.00 *ton*
Application event totals		116.62	28.53	181.22	5,428.16	

06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 12	Ground water	6.11	0.00	0.00	562.94	18,144,000.00 *gal*
Application event totals		6.11	0.00	0.00	562.94	

06/24/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
21-0-0	Solid commercial fertilizer	52.50	0.00	0.00	0.00	
Application event totals		52.50	0.00	0.00	0.00	

Field 14 - 06/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 12	Ground water	5.09	0.00	0.00	469.12	15,120,000.00 *gal*
Application event totals		5.09	0.00	0.00	469.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 12	Ground water	5.24	0.00	0.00	482.52	15,552,000.00 *gal*
Application event totals		5.24	0.00	0.00	482.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 12	Ground water	5.24	0.00	0.00	482.52	15,552,000.00 *gal*
Application event totals		5.24	0.00	0.00	482.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 12	Ground water	4.80	0.00	0.00	442.31	14,256,000.00 *gal*
Application event totals		4.80	0.00	0.00	442.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 12	Ground water	3.93	0.00	0.00	361.89	11,664,000.00 *gal*
Application event totals		3.93	0.00	0.00	361.89	

Field 17 - 11/10/2022: Oats, silage-soft dough

Field name: Field 17

Crop: Oats, silage-soft dough

Plant date: 11/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 17 - 11/10/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	123.72	9.70	15.29	0.00	1,000.00 *ton*
Application event totals		123.72	9.70	15.29	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	180.90	19,440,000.00 *gal*
Application event totals		0.00	0.00	0.00	180.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	211.05	22,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	211.05	

Field 17 - 05/14/2023: Corn, silage

Field name: Field 17

Crop: Corn, silage

Plant date: 05/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	211.05	22,680,000.00 *gal*
Application event totals		0.00	0.00	0.00	211.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Solid commercial fertilizer	105.00	0.00	0.00	0.00	
Application event totals		105.00	0.00	0.00	0.00	

Field 17 - 05/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	180.90	19,440,000.00 *gal*
Application event totals		0.00	0.00	0.00	180.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	180.90	19,440,000.00 *gal*
Application event totals		0.00	0.00	0.00	180.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Application event totals		64.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	150.75	16,200,000.00 *gal*
Application event totals		0.00	0.00	0.00	150.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	64.00	0.00	0.00	0.00	
Application event totals		64.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	150.75	16,200,000.00 *gal*
Application event totals		0.00	0.00	0.00	150.75	

Field 17 - 05/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	135.68	14,580,000.00 *gal*
Application event totals		0.00	0.00	0.00	135.68	

08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CCID	Ground water	0.00	0.00	0.00	150.75	16,200,000.00 *gal*
Application event totals		0.00	0.00	0.00	150.75	

Field 2 - 11/01/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	63.58	10.13	109.69	880.30	1,800,000.00 *gal*
IW 2	Ground water	12.84	0.00	0.00	615.69	9,000,000.00 *gal*
Application event totals		76.41	10.13	109.69	1,495.99	

03/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	70.42	10.82	108.78	892.88	1,620,000.00 *gal*
IW 2	Ground water	11.55	0.00	0.00	554.12	8,100,000.00 *gal*
Application event totals		81.98	10.82	108.78	1,446.99	

Field 2 - 05/04/2023: Corn, silage

Field 2 - 05/04/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 05/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	78.25	12.02	120.87	992.08	1,800,000.00 *gal*
IW 2	Ground water	12.84	0.00	0.00	615.69	9,000,000.00 *gal*
Application event totals		91.09	12.02	120.87	1,607.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	15.41	0.00	0.00	738.82	10,800,000.00 *gal*
Application event totals		15.41	0.00	0.00	738.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	78.57	7.63	141.34	1,486.73	1,680,000.00 *gal*
IW 2	Ground water	11.98	0.00	0.00	574.64	8,400,000.00 *gal*
Application event totals		90.56	7.63	141.34	2,061.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	14.55	0.00	0.00	697.78	10,200,000.00 *gal*
Application event totals		14.55	0.00	0.00	697.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	84.19	8.17	151.43	1,592.92	1,800,000.00 *gal*
IW 2	Ground water	12.84	0.00	0.00	615.69	9,000,000.00 *gal*
Application event totals		97.03	8.17	151.43	2,208.61	

Field 2 - 05/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	15.41	0.00	0.00	738.82	10,800,000.00 *gal*
Application event totals		15.41	0.00	0.00	738.82	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	17.97	0.00	0.00	861.96	12,600,000.00 *gal*
Application event totals		17.97	0.00	0.00	861.96	

Field 6 - 11/01/2022: Oats, silage-soft dough

Field name: Field 6

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.70	8.40	90.92	729.69	1,440,000.00 *gal*
IW 3	Ground water	7.36	0.00	0.00	539.45	7,776,000.00 *gal*
Application event totals		60.06	8.40	90.92	1,269.15	

03/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	81.08	12.45	125.24	1,027.94	1,800,000.00 *gal*
IW 3	Ground water	6.13	0.00	0.00	449.54	6,480,000.00 *gal*
Application event totals		87.21	12.45	125.24	1,477.49	

Field 6 - 04/28/2023: Corn, silage

Field 6 - 04/28/2023: Corn, silage

Field name: Field 6

Crop: Corn, silage

Plant date: 04/28/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	11.04	0.00	0.00	809.18	11,664,000.00 *gal*
Application event totals		11.04	0.00	0.00	809.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	9.81	0.00	0.00	719.27	10,368,000.00 *gal*
Application event totals		9.81	0.00	0.00	719.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	81.42	7.91	146.45	1,540.47	1,680,000.00 *gal*
IW 3	Ground water	8.58	0.00	0.00	629.36	9,072,000.00 *gal*
Application event totals		90.00	7.91	146.45	2,169.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	87.23	8.47	156.91	1,650.50	1,800,000.00 *gal*
IW 3	Ground water	9.81	0.00	0.00	719.27	10,368,000.00 *gal*
Application event totals		97.04	8.47	156.91	2,369.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	8.38	0.00	0.00	614.38	8,856,000.00 *gal*
Application event totals		8.38	0.00	0.00	614.38	

Field 6 - 04/28/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	8.58	0.00	0.00	629.36	9,072,000.00 *gal*
Application event totals		8.58	0.00	0.00	629.36	

Field 7 - 11/05/2022: Oats, silage-soft dough

Field name: Field 7

Crop: Oats, silage-soft dough

Plant date: 11/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/29/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	81.43	12.98	140.49	1,127.53	2,520,000.00 *gal*
IW 7	Ground water	10.67	0.00	0.00	429.54	7,560,000.00 *gal*
Application event totals		92.10	12.98	140.49	1,557.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	93.07	14.29	143.75	1,179.95	2,340,000.00 *gal*
IW 7	Ground water	9.91	0.00	0.00	398.86	7,020,000.00 *gal*
Application event totals		102.98	14.29	143.75	1,578.80	

Field 7 - 05/08/2023: Corn, silage

Field name: Field 7

Crop: Corn, silage

Plant date: 05/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 7 - 05/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	18.29	0.00	0.00	736.35	12,960,000.00 *gal*
Application event totals		18.29	0.00	0.00	736.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	13.72	0.00	0.00	552.26	9,720,000.00 *gal*
Application event totals		13.72	0.00	0.00	552.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	15.24	0.00	0.00	613.62	10,800,000.00 *gal*
Application event totals		15.24	0.00	0.00	613.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	61.62	5.98	110.84	1,165.89	1,440,000.00 *gal*
IW 7	Ground water	10.67	0.00	0.00	429.54	7,560,000.00 *gal*
Application event totals		72.29	5.98	110.84	1,595.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	92.43	8.97	166.25	1,748.83	2,160,000.00 *gal*
IW 7	Ground water	12.20	0.00	0.00	490.90	8,640,000.00 *gal*
Application event totals		104.62	8.97	166.25	2,239.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	15.24	0.00	0.00	613.62	10,800,000.00 *gal*
Application event totals		15.24	0.00	0.00	613.62	

Field 7 - 05/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	15.24	0.00	0.00	613.62	10,800,000.00 *gal*
Application event totals		15.24	0.00	0.00	613.62	

Field 8 - 11/01/2022: Oats, silage-soft dough

Field name: Field 8

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	138.04	21.20	213.22	1,750.14	960,000.00 *gal*
IW 7	Ground water	11.02	0.00	0.00	443.70	2,160,000.00 *gal*
Application event totals		149.06	21.20	213.22	2,193.84	

11/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	12.86	0.00	0.00	517.65	2,520,000.00 *gal*
Application event totals		12.86	0.00	0.00	517.65	

Field 8 - 05/02/2023: Corn, silage

Field name: Field 8

Crop: Corn, silage

Plant date: 05/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Field 8 - 05/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	120.78	18.55	186.57	1,531.37	840,000.00 *gal*
IW 7	Ground water	12.86	0.00	0.00	517.65	2,520,000.00 *gal*
Application event totals		133.64	18.55	186.57	2,049.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	16.53	0.00	0.00	665.55	3,240,000.00 *gal*
Application event totals		16.53	0.00	0.00	665.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	16.53	0.00	0.00	665.55	3,240,000.00 *gal*
Application event totals		16.53	0.00	0.00	665.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	14.70	0.00	0.00	591.60	2,880,000.00 *gal*
Application event totals		14.70	0.00	0.00	591.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	74.26	7.21	133.57	1,405.04	480,000.00 *gal*
Application event totals		74.26	7.21	133.57	1,405.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	11.02	0.00	0.00	443.70	2,160,000.00 *gal*
Application event totals		11.02	0.00	0.00	443.70	

Field 8 - 05/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	55.69	5.41	100.18	1,053.78	360,000.00 *gal*
IW 7	Ground water	11.02	0.00	0.00	443.70	2,160,000.00 *gal*
Application event totals		66.72	5.41	100.18	1,497.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	11.02	0.00	0.00	443.70	2,160,000.00 *gal*
Application event totals		11.02	0.00	0.00	443.70	

Field 9 - 10/28/2022: Oats, silage-soft dough

Field name: Field 9

Crop: Oats, silage-soft dough Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/11/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	109.35	17.42	188.66	1,514.12	2,160,000.00 *gal*
IW 9	Ground water	8.55	0.00	0.00	612.86	6,480,000.00 *gal*
Application event totals		117.91	17.42	188.66	2,126.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	112.16	17.22	173.24	1,421.99	1,800,000.00 *gal*
IW 9	Ground water	7.13	0.00	0.00	510.71	5,400,000.00 *gal*
Application event totals		119.28	17.22	173.24	1,932.70	

Field 9 - 06/05/2023: Corn, silage

Field 9 - 06/05/2023: Corn, silage

Field name: Field 9

Crop: Corn, silage

Plant date: 06/05/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	144.02	35.23	223.80	6,703.78	760.00 *ton*
Application event totals		144.02	35.23	223.80	6,703.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 9	Ground water	9.98	0.00	0.00	715.00	7,560,000.00 *gal*
Application event totals		9.98	0.00	0.00	715.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 9	Ground water	8.55	0.00	0.00	612.86	6,480,000.00 *gal*
Application event totals		8.55	0.00	0.00	612.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 9	Ground water	9.98	0.00	0.00	715.00	7,560,000.00 *gal*
Application event totals		9.98	0.00	0.00	715.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 9	Ground water	8.55	0.00	0.00	612.86	6,480,000.00 *gal*
Application event totals		8.55	0.00	0.00	612.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 9	Ground water	8.08	0.00	0.00	578.81	6,120,000.00 *gal*
Application event totals		8.08	0.00	0.00	578.81	

Field 9 - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	112.16	17.22	173.24	1,421.99	1,800,000.00 *gal*
Application event totals		112.16	17.22	173.24	1,421.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 9	Ground water	8.55	0.00	0.00	612.86	6,480,000.00 *gal*
Application event totals		8.55	0.00	0.00	612.86	

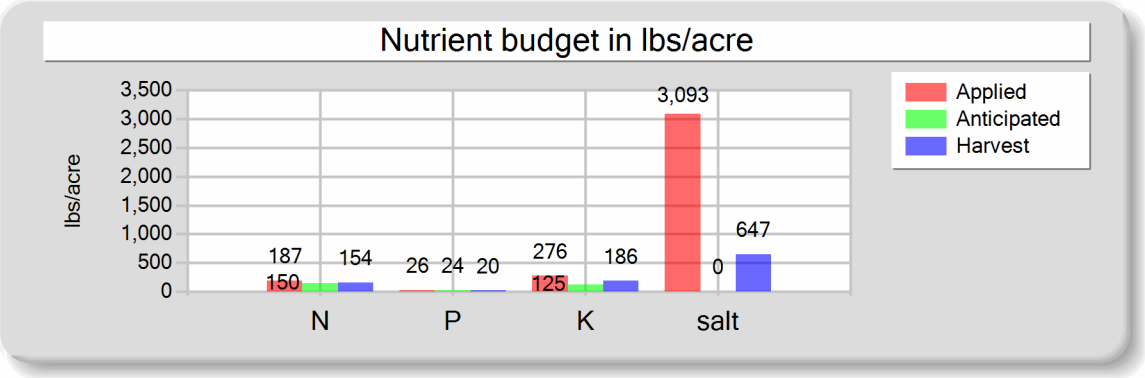
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 9	Ground water	8.55	0.00	0.00	612.86	6,480,000.00 *gal*
Application event totals		8.55	0.00	0.00	612.86	

B. NUTRIENT BUDGET

Field 10 - 10/27/2022: Oats, silage-soft dough

Field name: Field 10 Crop: Oats, silage-soft dough Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	168.24	26.34	275.67	2,235.12
Fresh water	11.97	0.00	0.00	858.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	187.21	26.34	275.67	3,093.12
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	154.49	19.56	185.59	647.06
Nutrient balance	32.72	6.78	90.07	2,446.06
Applied to removed ratio	1.21	1.35	1.49	4.78

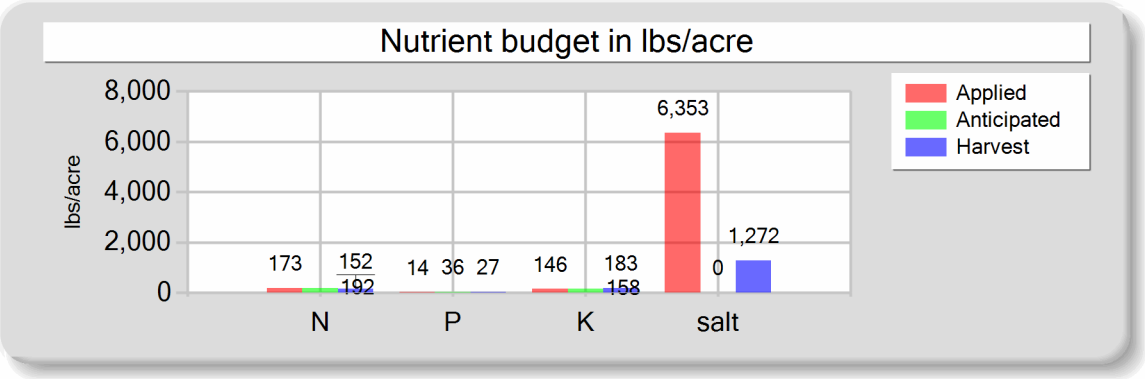
Fresh water applied	
15,120,000.00	*gallons*
556.82	*acre-inches*
5.57	*inches/acre*

Process wastewater applied	
5,040,000.00	*gallons*
185.61	*acre-inches*
1.86	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 10 - 05/19/2023: Corn, silage

Field name: Field 10 Crop: Corn, silage Plant date: 05/19/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	94.21	14.47	145.52	1,194.47
Fresh water	71.99	0.00	0.00	5,158.21
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	173.20	14.47	145.52	6,352.68
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	152.31	26.77	183.22	1,271.77
Nutrient balance	20.89	-12.30	-37.70	5,080.91
Applied to removed ratio	1.14	0.54	0.79	5.00

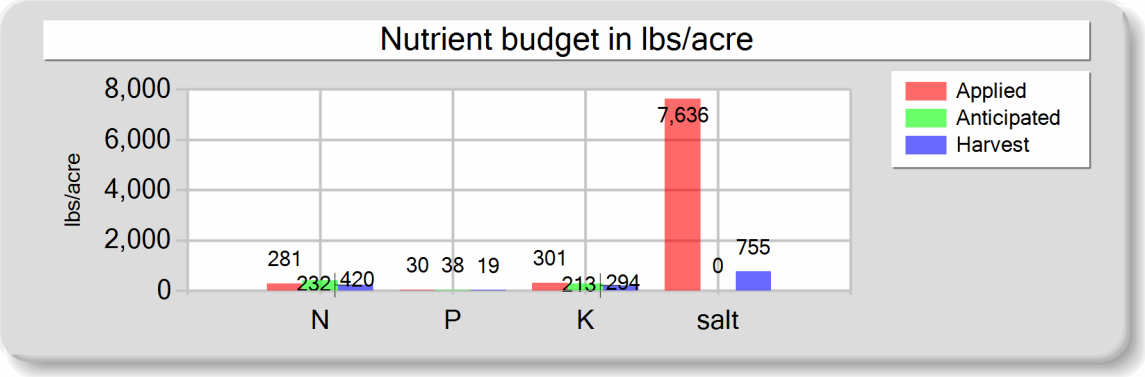
Fresh water applied	
90,900,000.00	*gallons*
3,347.54	*acre-inches*
33.48	*inches/acre*

Process wastewater applied	
2,520,000.00	*gallons*
92.80	*acre-inches*
0.93	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 11 - 01/01/2023: Alfalfa, hay

Field name: Field 11 Crop: Alfalfa, hay Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	194.59	29.88	300.56	2,467.06
Fresh water	72.13	0.00	0.00	5,168.67
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	280.72	29.88	300.56	7,635.74
Anticipated crop nutrient removal	420.00	37.80	294.00	0.00
Actual crop nutrient removal	231.56	19.11	212.83	755.39
Nutrient balance	49.15	10.77	87.74	6,880.34
Applied to removed ratio	1.21	1.56	1.41	10.11

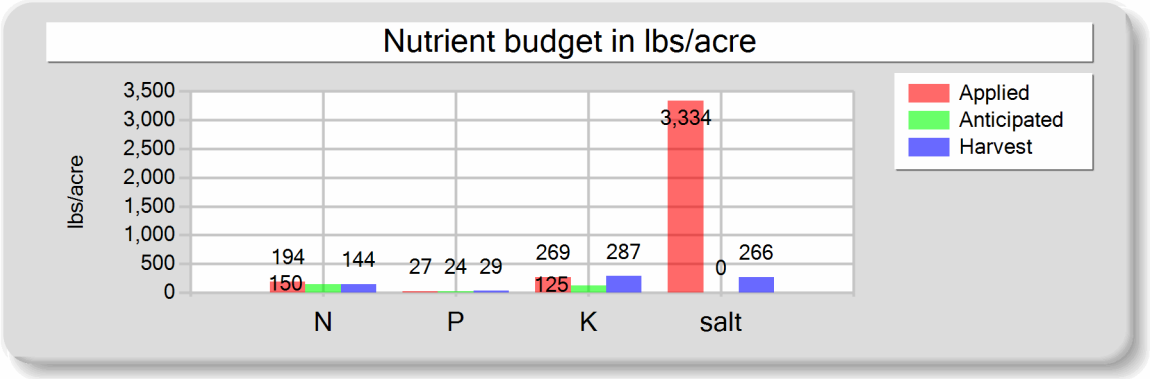
Fresh water applied	
75,600,000.00	*gallons*
2,784.09	*acre-inches*
33.54	*inches/acre*

Process wastewater applied	
4,320,000.00	*gallons*
159.09	*acre-inches*
1.92	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 12 - 11/03/2022: Oats, silage-soft dough

Field name: Field 12 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	174.47	26.79	269.49	2,211.98
Fresh water	12.17	0.00	0.00	1,121.69
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	193.64	26.79	269.49	3,333.67
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	143.65	29.29	287.30	265.89
Nutrient balance	49.99	-2.50	-17.81	3,067.78
Applied to removed ratio	1.35	0.91	0.94	12.54

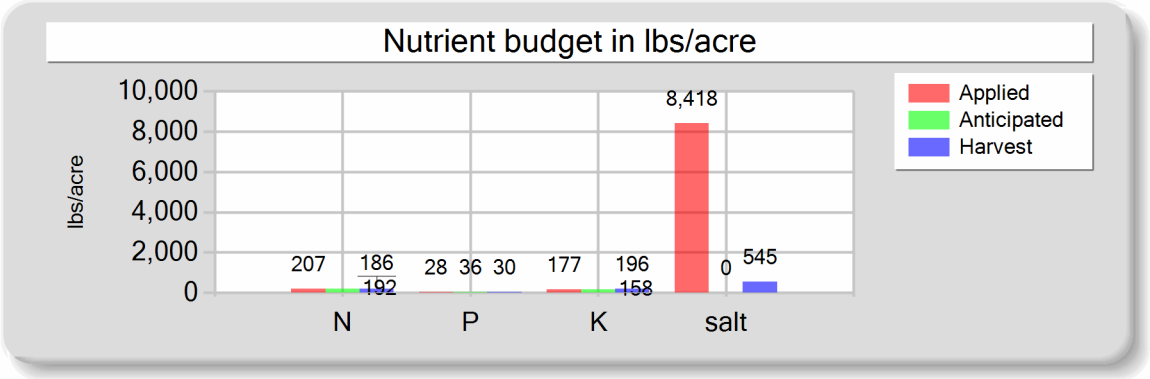
Fresh water applied	
22,248,000.00	*gallons*
819.32	*acre-inches*
11.38	*inches/acre*

Process wastewater applied	
3,360,000.00	*gallons*
123.74	*acre-inches*
1.72	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 12 - 06/08/2023: Corn, silage

Field name: Field 12 Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	52.50	0.00	0.00	0.00
Dry manure	113.70	27.81	176.69	5,292.46
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	33.91	0.00	0.00	3,125.49
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	207.12	27.81	176.69	8,417.95
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	185.59	29.54	196.17	544.78
Nutrient balance	21.52	-1.72	-19.48	7,873.17
Applied to removed ratio	1.12	0.94	0.90	15.45

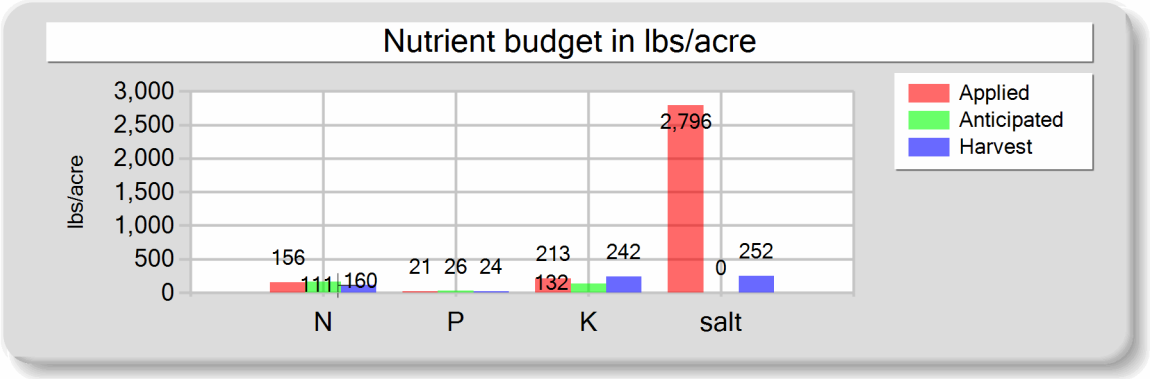
Fresh water applied	
61,992,000.00	*gallons*
2,282.95	*acre-inches*
31.71	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 14 - 11/04/2022: Oats, hay

Field name: Field 14 Crop: Oats, hay Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	138.04	21.20	213.22	1,750.14
Fresh water	11.34	0.00	0.00	1,045.46
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	156.38	21.20	213.22	2,795.60
Anticipated crop nutrient removal	160.00	26.00	132.00	0.00
Actual crop nutrient removal	110.68	23.91	241.73	251.83
Nutrient balance	45.70	-2.71	-28.51	2,543.77
Applied to removed ratio	1.41	0.89	0.88	11.10

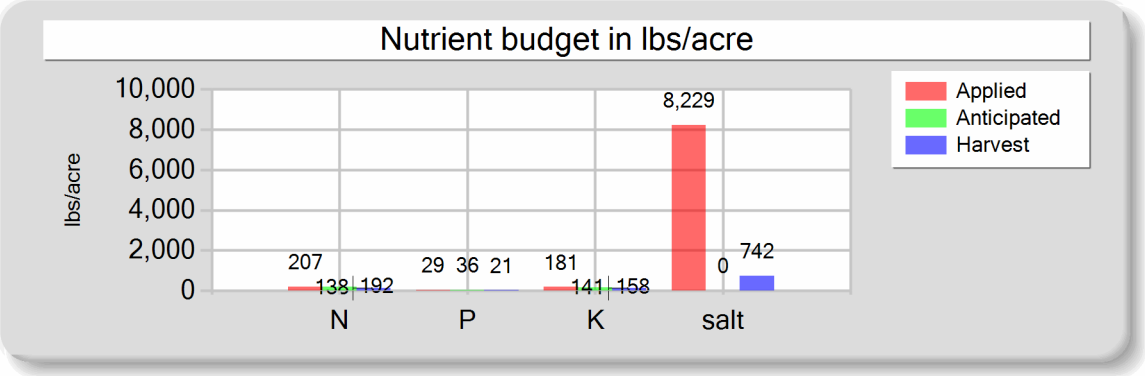
Fresh water applied	
33,696,000.00	*gallons*
1,240.91	*acre-inches*
10.61	*inches/acre*

Process wastewater applied	
4,320,000.00	*gallons*
159.09	*acre-inches*
1.36	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 14 - 06/24/2023: Corn, silage

Field name: Field 14 Crop: Corn, silage Plant date: 06/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	52.50	0.00	0.00	0.00
Dry manure	116.62	28.53	181.22	5,428.16
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	30.40	0.00	0.00	2,801.30
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	206.51	28.53	181.22	8,229.47
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	138.35	21.09	140.57	742.20
Nutrient balance	68.17	7.44	40.65	7,487.27
Applied to removed ratio	1.49	1.35	1.29	11.09

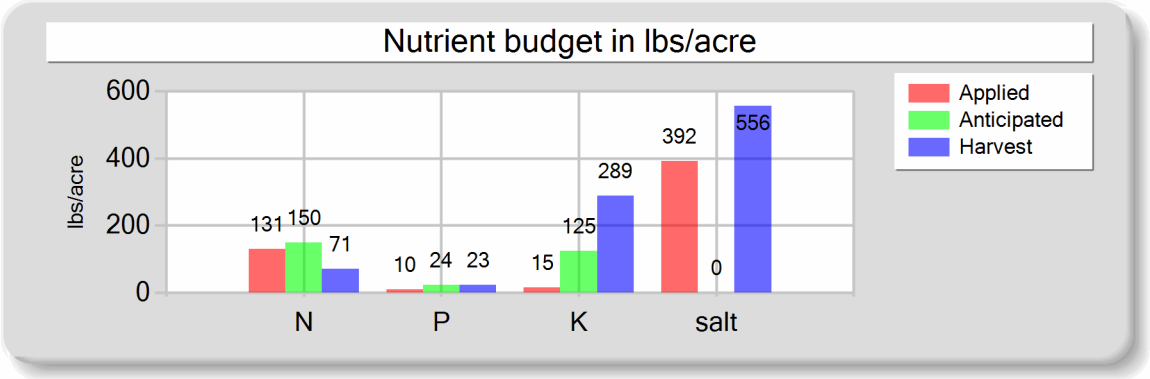
Fresh water applied	
90,288,000.00	*gallons*
3,325.00	*acre-inches*
28.42	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 17 - 11/10/2022: Oats, silage-soft dough

Field name: Field 17 Crop: Oats, silage-soft dough Plant date: 11/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	123.72	9.70	15.29	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	391.95
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	130.72	9.70	15.29	391.95
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	70.94	22.90	288.59	555.77
Nutrient balance	59.79	-13.20	-273.30	-163.82
Applied to removed ratio	1.84	0.42	0.05	0.71

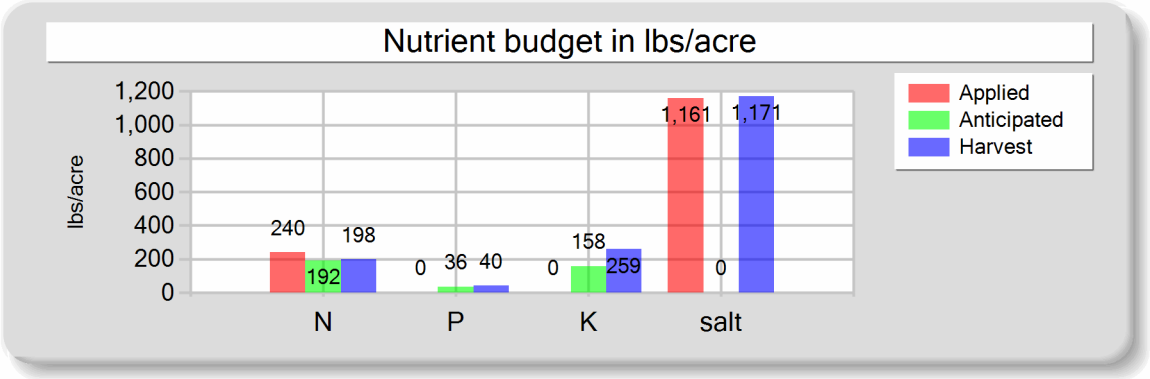
Fresh water applied	
42,120,000.00	*gallons*
1,551.14	*acre-inches*
11.16	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 17 - 05/14/2023: Corn, silage

Field name: Field 17 Crop: Corn, silage Plant date: 05/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	233.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,160.78
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	240.00	0.00	0.00	1,160.78
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	197.89	40.48	258.85	1,170.83
Nutrient balance	42.11	-40.48	-258.85	-10.05
Applied to removed ratio	1.21	0.00	0.00	0.99

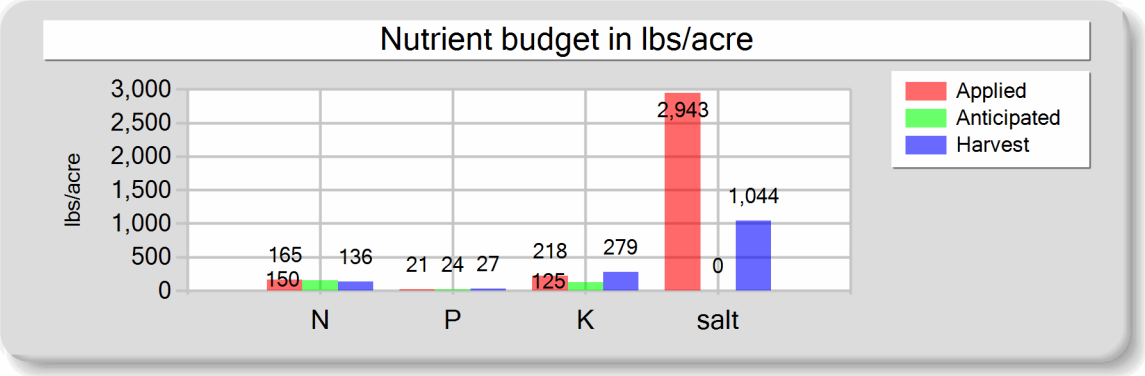
Fresh water applied	
124,740,000.00	*gallons*
4,593.75	*acre-inches*
33.05	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 11/01/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	134.00	20.95	218.47	1,773.18
Fresh water	24.39	0.00	0.00	1,169.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	165.39	20.95	218.47	2,942.98
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	136.48	26.88	279.17	1,044.29
Nutrient balance	28.91	-5.94	-60.70	1,898.69
Applied to removed ratio	1.21	0.78	0.78	2.82

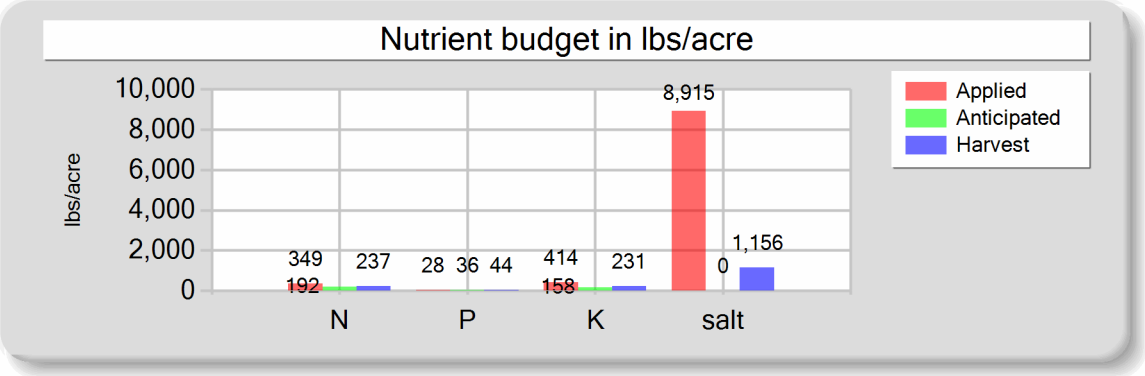
Fresh water applied	
17,100,000.00	*gallons*
629.73	*acre-inches*
7.32	*inches/acre*

Process wastewater applied	
3,420,000.00	*gallons*
125.95	*acre-inches*
1.46	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 05/04/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 05/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	241.01	27.82	413.64	4,071.74
Fresh water	100.99	0.00	0.00	4,843.40
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	349.00	27.82	413.64	8,915.14
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	236.58	44.00	230.88	1,156.28
Nutrient balance	112.42	-16.18	182.75	7,758.86
Applied to removed ratio	1.48	0.63	1.79	7.71

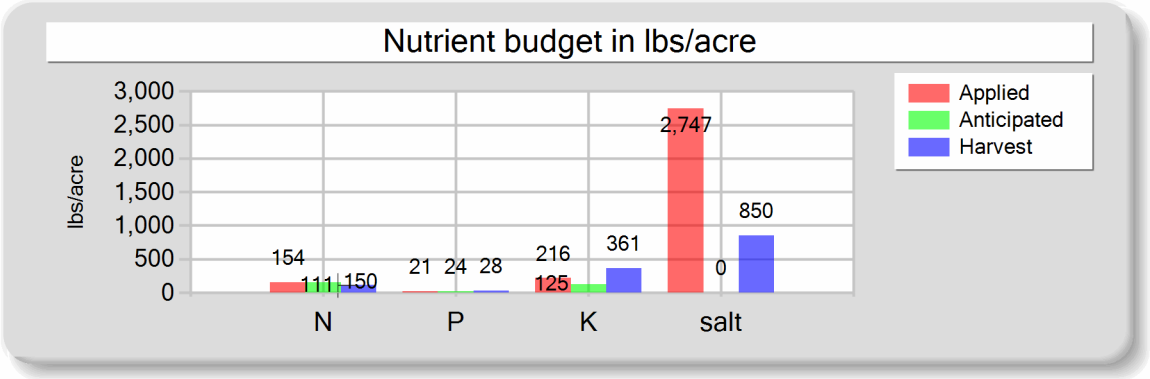
Fresh water applied	
70,800,000.00	*gallons*
2,607.32	*acre-inches*
30.32	*inches/acre*

Process wastewater applied	
5,280,000.00	*gallons*
194.44	*acre-inches*
2.26	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6 - 11/01/2022: Oats, silage-soft dough

Field name: Field 6 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	133.78	20.85	216.16	1,757.64
Fresh water	13.49	0.00	0.00	989.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	154.27	20.85	216.16	2,746.64
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	110.86	27.72	361.33	849.95
Nutrient balance	43.40	-6.87	-145.17	1,896.69
Applied to removed ratio	1.39	0.75	0.60	3.23

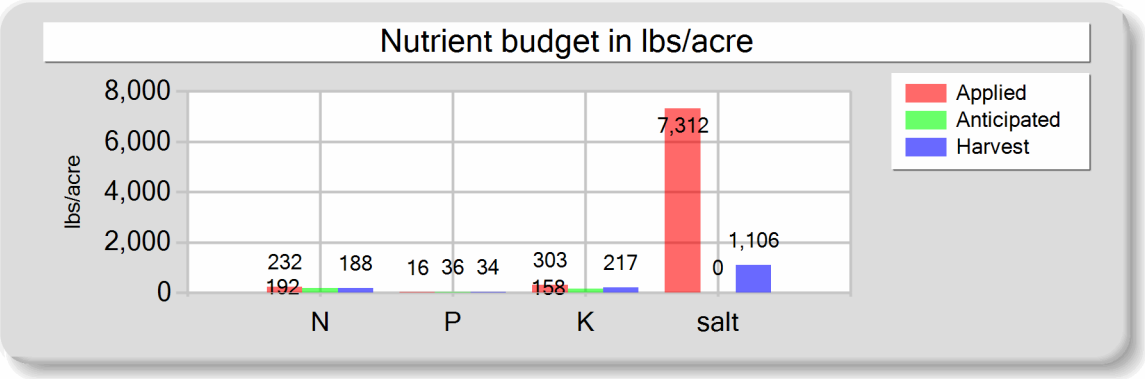
Fresh water applied	
14,256,000.00	*gallons*
525.00	*acre-inches*
6.33	*inches/acre*

Process wastewater applied	
3,240,000.00	*gallons*
119.32	*acre-inches*
1.44	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 6 - 04/28/2023: Corn, silage

Field name: Field 6 Crop: Corn, silage Plant date: 04/28/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	168.65	16.37	303.35	3,190.97
Fresh water	56.20	0.00	0.00	4,120.82
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	231.84	16.37	303.35	7,311.79
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	188.50	33.96	217.26	1,105.75
Nutrient balance	43.34	-17.58	86.09	6,206.04
Applied to removed ratio	1.23	0.48	1.40	6.61

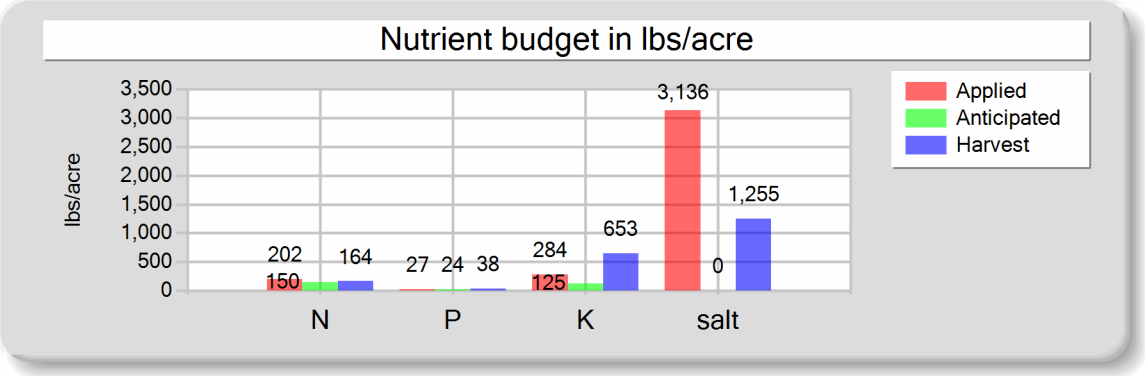
Fresh water applied	
59,400,000.00	*gallons*
2,187.50	*acre-inches*
26.36	*inches/acre*

Process wastewater applied	
3,480,000.00	*gallons*
128.16	*acre-inches*
1.54	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 7 - 11/05/2022: Oats, silage-soft dough

Field name: Field 7 Crop: Oats, silage-soft dough Plant date: 11/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	174.50	27.27	284.25	2,307.48
Fresh water	20.58	0.00	0.00	828.39
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	202.08	27.27	284.25	3,135.87
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	163.72	38.26	652.51	1,254.60
Nutrient balance	38.36	-10.99	-368.26	1,881.27
Applied to removed ratio	1.23	0.71	0.44	2.50

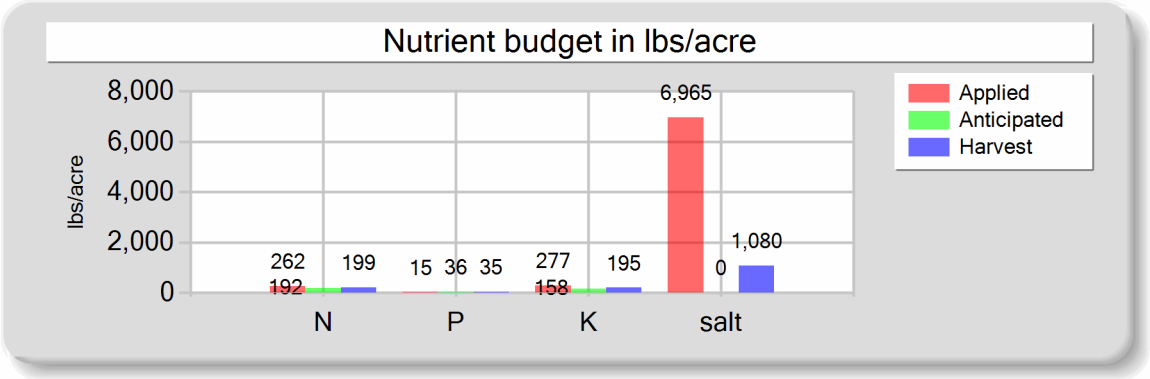
Fresh water applied	
14,580,000.00	*gallons*
536.93	*acre-inches*
5.71	*inches/acre*

Process wastewater applied	
4,860,000.00	*gallons*
178.98	*acre-inches*
1.90	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 7 - 05/08/2023: Corn, silage

Field name: Field 7 Crop: Corn, silage Plant date: 05/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	154.05	14.96	277.09	2,914.71
Fresh water	100.62	0.00	0.00	4,049.92
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	261.66	14.96	277.09	6,964.63
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	198.62	35.05	194.90	1,079.55
Nutrient balance	63.04	-20.09	82.19	5,885.08
Applied to removed ratio	1.32	0.43	1.42	6.45

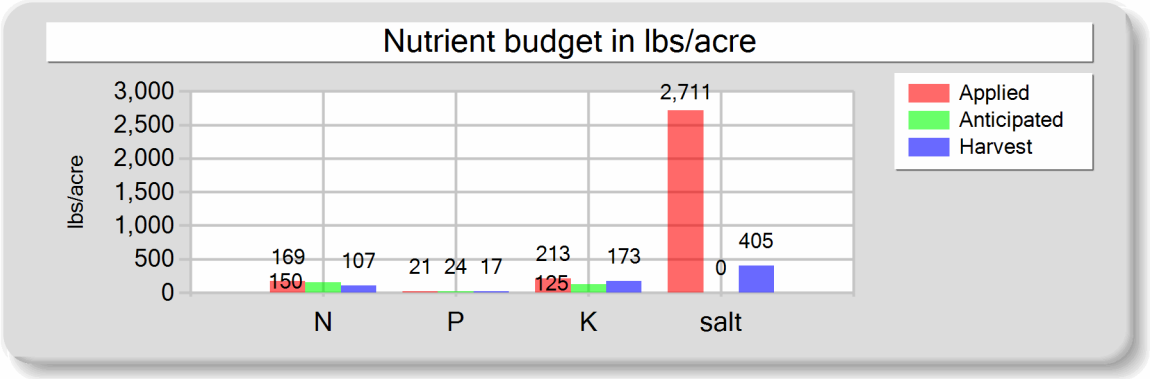
Fresh water applied	
71,280,000.00	*gallons*
2,625.00	*acre-inches*
27.93	*inches/acre*

Process wastewater applied	
3,600,000.00	*gallons*
132.58	*acre-inches*
1.41	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 8 - 11/01/2022: Oats, silage-soft dough

Field name: Field 8 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	138.04	21.20	213.22	1,750.14
Fresh water	23.88	0.00	0.00	961.34
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	168.92	21.20	213.22	2,711.48
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	106.67	17.04	172.85	405.20
Nutrient balance	62.25	4.16	40.38	2,306.28
Applied to removed ratio	1.58	1.24	1.23	6.69

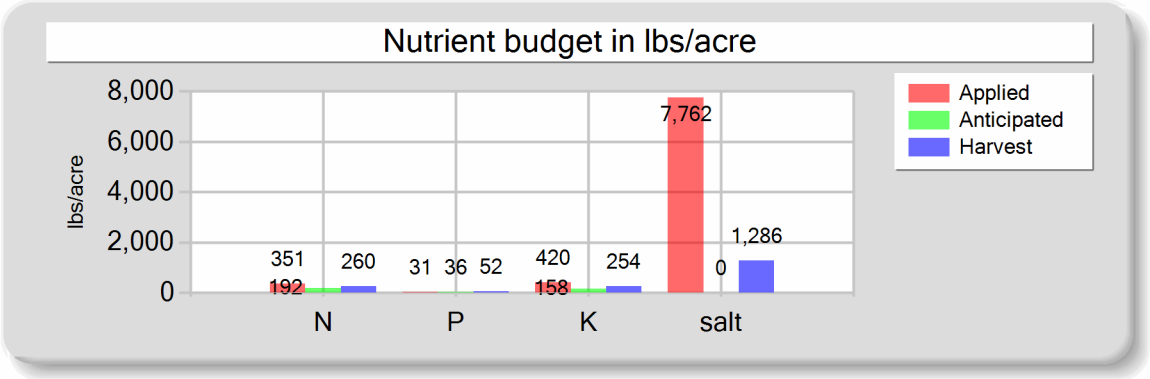
Fresh water applied	
4,680,000.00	*gallons*
172.35	*acre-inches*
6.63	*inches/acre*

Process wastewater applied	
960,000.00	*gallons*
35.35	*acre-inches*
1.36	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 8 - 05/02/2023: Corn, silage

Field name: Field 8 Crop: Corn, silage Plant date: 05/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	250.74	31.17	420.32	3,990.19
Fresh water	93.70	0.00	0.00	3,771.43
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	351.43	31.17	420.32	7,761.62
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	259.77	51.84	254.26	1,286.38
Nutrient balance	91.66	-20.68	166.06	6,475.24
Applied to removed ratio	1.35	0.60	1.65	6.03

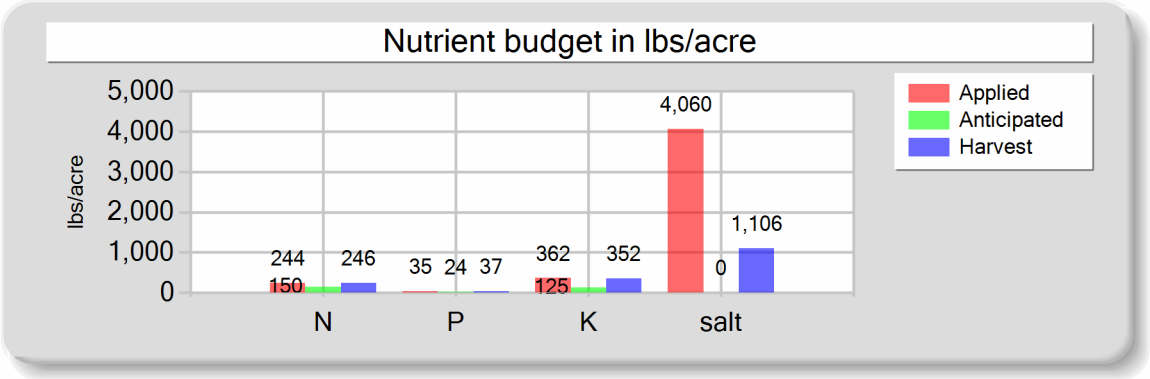
Fresh water applied	
18,360,000.00	*gallons*
676.14	*acre-inches*
26.01	*inches/acre*

Process wastewater applied	
1,680,000.00	*gallons*
61.87	*acre-inches*
2.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 9 - 10/28/2022: Oats, silage-soft dough

Field name: Field 9 Crop: Oats, silage-soft dough Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	221.51	34.65	361.91	2,936.10
Fresh water	15.68	0.00	0.00	1,123.57
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	244.19	34.65	361.91	4,059.68
Anticipated crop nutrient removal	150.00	24.00	124.50	0.00
Actual crop nutrient removal	245.98	37.30	352.09	1,105.56
Nutrient balance	-1.79	-2.65	9.81	2,954.12
Applied to removed ratio	0.99	0.93	1.03	3.67

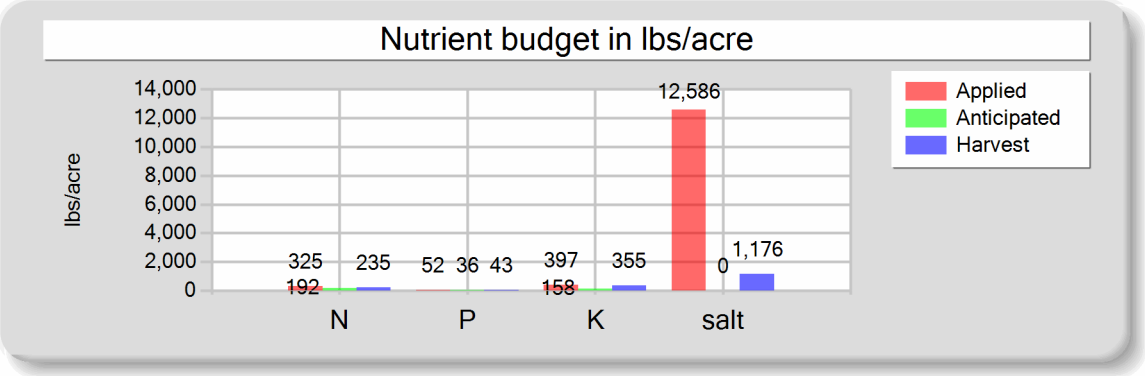
Fresh water applied	
11,880,000.00	*gallons*
437.50	*acre-inches*
7.29	*inches/acre*

Process wastewater applied	
3,960,000.00	*gallons*
145.83	*acre-inches*
2.43	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 9 - 06/05/2023: Corn, silage

Field name: Field 9 Crop: Corn, silage Plant date: 06/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	144.02	35.23	223.80	6,703.78
Process wastewater	112.16	17.22	173.24	1,421.99
Fresh water	62.25	0.00	0.00	4,460.24
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	325.43	52.45	397.05	12,586.01
Anticipated crop nutrient removal	192.00	36.00	158.40	0.00
Actual crop nutrient removal	234.88	43.22	354.50	1,175.88
Nutrient balance	90.55	9.24	42.54	11,410.13
Applied to removed ratio	1.39	1.21	1.12	10.70

Fresh water applied	
47,160,000.00	*gallons*
1,736.74	*acre-inches*
28.95	*inches/acre*

Process wastewater applied	
1,800,000.00	*gallons*
66.29	*acre-inches*
1.10	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Fall SM

Sample and source description: Fall SM

Sample date: 10/21/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,900.00	2,030.00	3,200.00							0.00
DL	100.00	100.00	100.00							0.01

Spring SM

Sample and source description: Spring SM

Sample date: 05/03/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,900.00	3,400.00	21,600.00							64.70
DL	100.00	100.00	100.00							0.01

Fall SM

Sample and source description: Fall SM

Sample date: 10/05/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,100.00	3,050.00	4,840.00							67.70
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

4QWW

Sample and source description: 4QWW

Sample date: 12/21/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	364.00	250.00	0.00	0.00	58.00	628.00								6,080.00	5,040
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

1QWW

Sample and source description: 1QWW

Sample date: 03/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	448.00	272.00	0.00	0.00	68.80	692.00								6,830.00	5,680
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

2QWW

Sample and source description: 2QWW

Sample date: 05/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	482.00	353.00	0.00	0.00	46.80	867.00								7,830.00	9,120
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

3QWW

Sample and source description: 3QWW

Sample date: 10/02/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	258.00	193.00	0.00	0.00	60.80	488.00								5,320.00	3,960
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

4QWW

Sample and source description: 4QWW

Sample date: 12/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	501.00	325.00	0.00	0.00	57.90	630.00								6,110.00	6,880
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

CCID

Fresh Water

Sample description: Fresh Water

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								230.00	155
DL	0.50	0.50	0.50								1.00	10

IW 12

Fresh Water

Sample description: Fresh Water

Sample date: 01/03/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.72	0.00	4.72								930.00	435
DL	0.50	0.50	0.50								1.00	10

IW 2

IW 2

Fresh Water

Sample description: Fresh Water

Sample date: 01/03/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	14.70	0.00	14.70								1,360.00	705
DL	0.50	0.50	0.50								1.00	10

IW 7

Fresh Water

Sample description: Fresh Water

Sample date: 01/03/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	15.90	0.00	15.90								1.31	640
DL	0.50	0.50	0.50								1.00	10

IW 8

Fresh Water

Sample description: Fresh Water

Sample date: 01/03/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	26.40	0.00	23.60								1,800.00	995
DL	0.50	0.50	0.50								1.00	10

IW 9

IW 9

Fresh Water

Sample description: Fresh Water

Sample date: 01/03/2024 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	9.49	0.00	9.46								1,300.00	680
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field 10 - 10/27/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,160.00	780.00	7,400.00		8.60
DL	100.00	100.00	100.00		0.01

Field 10 - 05/19/2023: Corn, silage

Field 10 - 05/19/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,040.00	710.00	4,860.00		10.10
DL	100.00	100.00	100.00		0.01

Field 11 - 01/01/2023: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 06/16/2022 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 10.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	30,900.00	2,550.00	28,400.00		11.20
DL	100.00	100.00	100.00		0.01

Field 12 - 11/03/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,100.00	1,040.00	10,200.00		9.44
DL	100.00	100.00	100.00		0.01

Field 12 - 06/08/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,210.00	670.00	4,450.00		3.70
DL	100.00	100.00	100.00		0.01

Field 14 - 11/04/2022: Oats, hay

Forage

Sample and source description: Forage

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,170.00	900.00	9,060.00		9.48
DL	100.00	100.00	100.00		0.01

Field 14 - 06/24/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,740.00	570.00	3,800.00		6.60
DL	100.00	100.00	100.00		0.01

Field 17 - 11/10/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/20/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,810.00	1,230.00	15,500.00		9.95
DL	100.00	100.00	100.00		0.01

Forage

Sample and source description: Forage

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,710.00	910.00	9,720.00		8.78
DL	100.00	100.00	100.00		0.01

Field 17 - 05/14/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,960.00	810.00	5,180.00		7.10
DL	100.00	100.00	100.00		0.01

Field 2 - 11/01/2022: Oats, silage-soft dough

Field 2 - 11/01/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,960.00	780.00	8,100.00		10.10
DL	100.00	100.00	100.00		0.01

Field 2 - 05/04/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,570.00	850.00	4,460.00		6.40
DL	100.00	100.00	100.00		0.01

Field 6 - 04/28/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 52.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,440.00	980.00	6,270.00		6.69
DL	100.00	100.00	100.00		0.01

Field 7 - 11/05/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/20/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,520.00	1,290.00	22,000.00		14.10
DL	100.00	100.00	100.00		0.01

Field 7 - 05/08/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,740.00	660.00	3,670.00		5.60
DL	100.00	100.00	100.00		0.01

Field 8 - 11/01/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 09/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,320.00	690.00	7,000.00		5.47
DL	100.00	100.00	100.00		0.01

Field 8 - 05/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/05/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,710.00	940.00	4,610.00		6.39
DL	100.00	100.00	100.00		0.01

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,050.00	900.00	4,640.00		8.20
DL	100.00	100.00	100.00		0.01

Field 9 - 10/28/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,540.00	840.00	7,930.00		8.30
DL	100.00	100.00	100.00		0.01

Field 9 - 06/05/2023: Corn, silage

Field 9 - 06/05/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,750.00	690.00	5,660.00		6.30
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

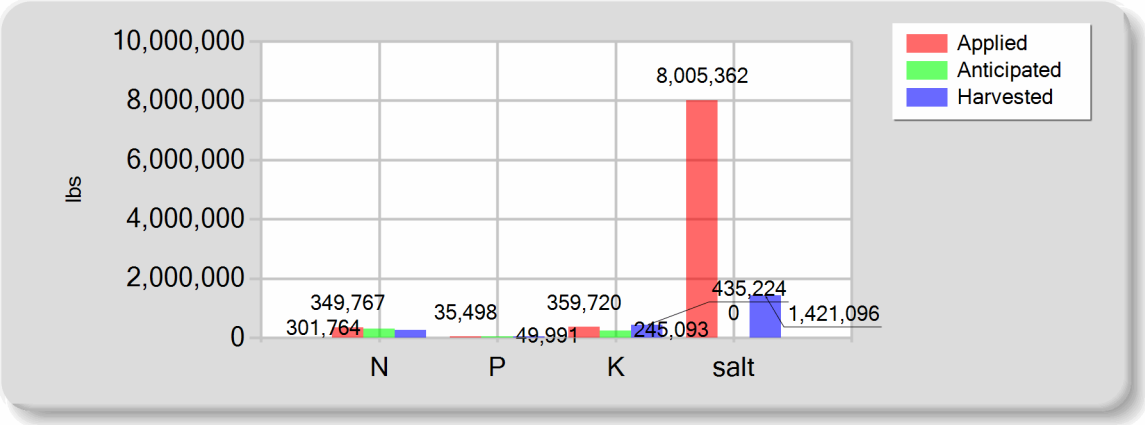
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

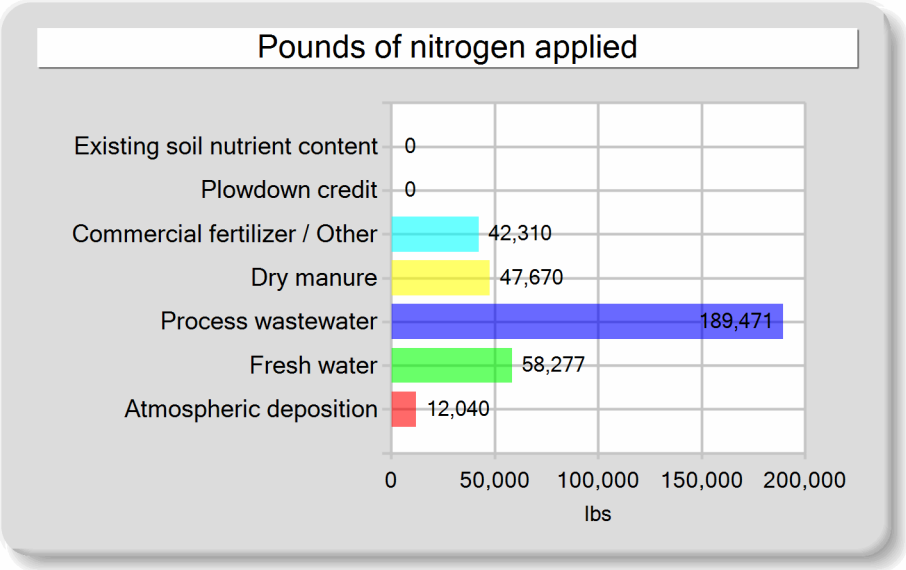
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

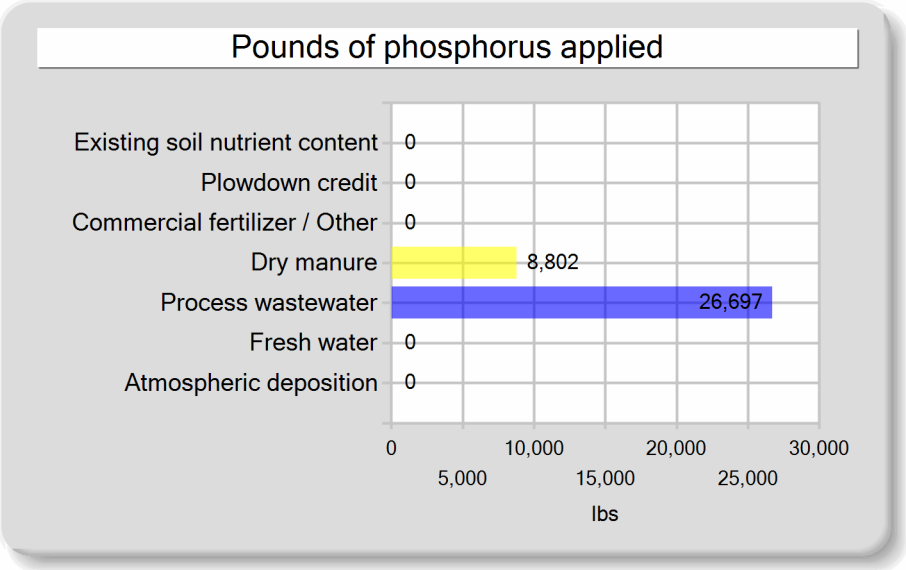
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	42,309.50	0.00	0.00	0.00
Dry manure	47,669.74	8,801.54	49,477.18	1,418,379.28
Process wastewater	189,470.89	26,696.92	310,242.73	2,726,772.14
Fresh water	58,276.67	0.00	0.00	3,860,210.74
Atmospheric deposition	12,040.00	0.00	0.00	0.00
Total nutrients applied	349,766.79	35,498.46	359,719.92	8,005,362.16
Anticipated crop nutrient removal	301,764.00	49,991.40	245,092.80	0.00
Actual crop nutrient removal	269,476.98	49,256.36	435,224.00	1,421,095.96
Nutrient balance	80,289.81	-13,757.90	-75,504.08	6,584,266.20
Applied to removed ratio	1.30	0.72	0.83	5.63

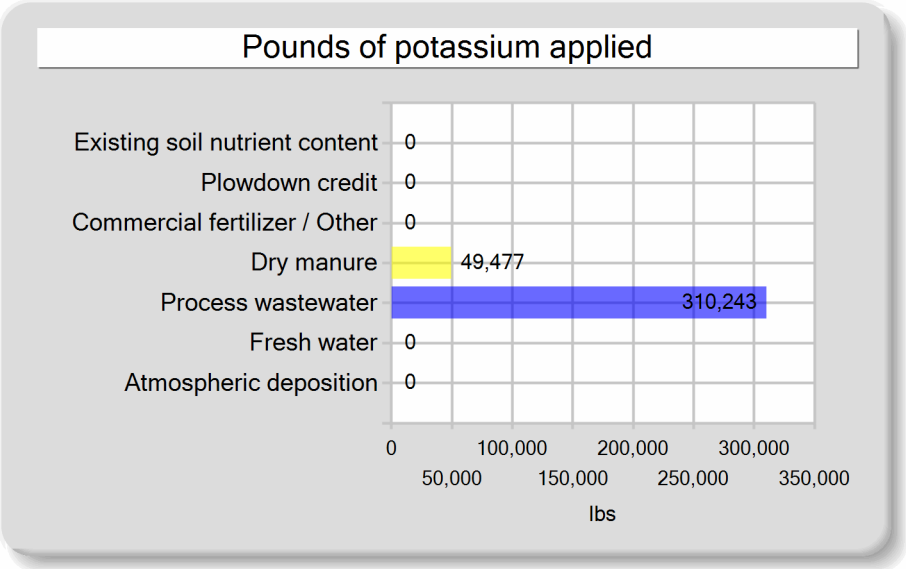
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

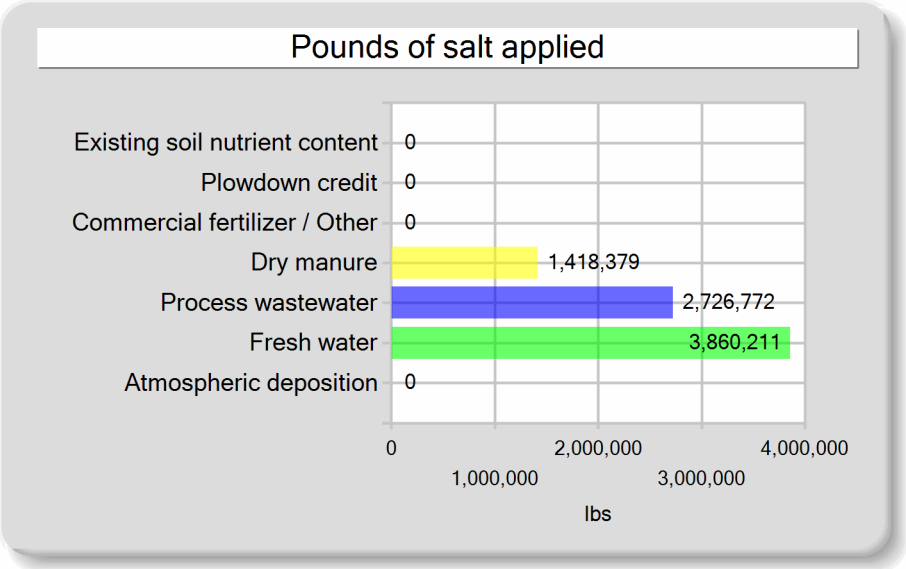


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Poor forage yields on Field 8 and Field 17 resulted in nitrogen application to removal ratios exceeding 1.50 and 1.80 respectively.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Paul Lopes
Paul Lopes (Apr 18, 2024 23:47 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Paul Lopes	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Apr 18, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Paul Lopes

Name of Dairy Facility: Tony Lopes Dairy LP

Facility Address:

Number and Street	City	County	Zip Code
27500 W Bunker RD	Gustine	Merced	95322

Contact Person Name and Phone Number: Paul Lopes (209) 649-4967
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: J&F Fertilizer

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
900 Meredith Ave	Gustine	CA	95322

Contact Person: Office (209) 854-6325
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

P&D Dairy North (209) 649-4967
Name Phone Number

Address	City	State	Zip Code
28366 Husman Rd.	Gustine	CA	95322

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code
28366 Husman Rd	Gustine	95322

Street and nearest cross street (if no address) — County: Merced

Assessor's Parcel Number — Assessor's Parcel Number County

Last date hauled: 05/17/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 880.00 tons

Manure Solids Content: 40.9 %

Method used to determine amount of manure:

Haulers Calculations

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Paul Lopes
Paul Lopes (Apr 18, 2024 23:47 PDT)

Apr 18, 2024

Operator Signature | Date

John Nunes

Hauler Signature | Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

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OPERATOR INFORMATION

Name of Operator: Paul Lopes

Name of Dairy Facility: Tony Lopes Dairy LP

Facility Address:

Number and Street	City	County	Zip Code
27500 W Bunker RD	Gustine	Merced	95322

Contact Person Name and Phone Number: Paul Lopes (209) 649-4967
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: J&F Fertilizer

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
900 Meredith Ave	Gustine	CA	95322

Contact Person: Office (209) 854-6325
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

P&D Dairy North (209) 649-4967
Name Phone Number

Address	City	State	Zip Code
28366 Husman Rd.	Gustine	CA	95322

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code
28366 Husman Rd	Gustine	95322

Street and nearest cross street (if no address) — County: Merced

Assessor's Parcel Number — Assessor's Parcel Number County

Last date hauled: 05/17/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 1,300.00 tons

Manure Solids Content: 40.9 %

Method used to determine amount of manure:

Haulers Calculations

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Paul Lopes
Paul Lopes (Apr 18, 2024 23:47 PDT)

Apr 18, 2024

Operator Signature | Date

John Nunes

Hauler Signature | Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

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Name of Dairy Facility: Tony Lopes Dairy LP

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Number and Street	City	County	Zip Code
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Contact Person Name and Phone Number: Paul Lopes (209) 649-4967
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: J&F Fertilizer

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
900 Meredith Ave	Gustine	CA	95322

Contact Person: Office (209) 854-6325
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

P&D Dairy North (209) 649-4967
Name Phone Number

Address	City	State	Zip Code
28366 Husman Rd.	Gustine	CA	95322

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code
28366 Husman Rd	Gustine	95322

Street and nearest cross street (if no address) — County: Merced

Assessor's Parcel Number — Assessor's Parcel Number County

Last date hauled: 10/19/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 270.00 tons

Manure Solids Content: 41.3 %

Method used to determine amount of manure:

Haulers Calculations

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Paul Lopes — Apr 18, 2024

Paul Lopes (Apr 18, 2024 23:47 PDT)

Operator Signature — Date

John Nunes

Hauler Signature — Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Paul Lopes

Name of Dairy Facility: Tony Lopes Dairy LP

Facility Address:

Number and Street	City	County	Zip Code
27500 W Bunker RD	Gustine	Merced	95322

Contact Person Name and Phone Number: Paul Lopes (209) 649-4967
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: J&F Fertilizer

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
900 Meredith Ave	Gustine	CA	95322

Contact Person: Office (209) 854-6325
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

P&D Dairy North (209) 649-4967
Name Phone Number

Address	City	State	Zip Code
28366 Husman Rd.	Gustine	CA	95322

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code
28366 Husman Rd	Gustine	95322

Street and nearest cross street (if no address) — County: Merced

Assessor's Parcel Number — Assessor's Parcel Number County

Last date hauled: 10/19/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 820.00 tons

Manure Solids Content: 41.3 %

Method used to determine amount of manure:

Haulers Calculations

CERTIFICATION

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Paul Lopes
Paul Lopes (Apr 18, 2024 23:47 PDT)

Apr 18, 2024

Operator Signature | Date

John Nunes

Hauler Signature | Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

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2) If there are multiple destinations, complete a separate form for each destination.
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4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Paul Lopes

Name of Dairy Facility: Tony Lopes Dairy LP

Facility Address:

Number and Street	City	County	Zip Code
27500 W Bunker RD	Gustine	Merced	95322

Contact Person Name and Phone Number: Paul Lopes (209) 649-4967
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: J&F Fertilizer

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
900 Meredith Ave	Gustine	CA	95322

Contact Person: Office (209) 854-6325
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Farmer

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

P&D Dairy North (209) 649-4967
Name Phone Number

Address	City	State	Zip Code
28366 Husman Rd.	Gustine	CA	95322

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code
28366 Husman Rd	Gustine	95322

Street and nearest cross street (if no address) — County: Merced

Assessor's Parcel Number — Assessor's Parcel Number County

Last date hauled: 10/19/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 430.00 tons

Manure Solids Content: 41.3 %

Method used to determine amount of manure:

Haulers Calculations

CERTIFICATION

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Paul Lopes
Paul Lopes (Apr 18, 2024 23:47 PDT)

Apr 18, 2024

Operator Signature | Date

John Nunes

Hauler Signature | Date

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
TONY L. LOPES DAIRY L.P.
9270 S. Whitworth Road
Gustine, CA 95322

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 1/3/24 9:10 AM
SUBMITTED BY: Joe Ramos
ANALYZED DATE: 1/9/24

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 11.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW 2	T4003007A	Electrical Conductivity (EC)	EPA 120.1 and	1.36	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	14.7	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	705	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	14.7	mg/L
		Soluble Salts (SALT-SOL)	Calculation	868	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	99.8	97.0 (T4003004A)	96.4 (T4003004A)			0.619
Electrical Conductivity (EC)	99.2			1.54 (T4003002B)	1.52 (Dup)	1.05
Nitrate Nitrogen (NO3-N)	101	97.9 (T4003004B)	95.9 (T4003004B)			2.04
Total Dissolved Solids (TDS)	103			435 (T4003007F)	440 (Dup)	1.14
Total Nitrogen (TN)	104			0.000 (T4003007A)	0.000 (null)	

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
TONY L. LOPES DAIRY L.P.
9270 S. Whitworth Road
Gustine, CA 95322

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 1/3/24 9:10 AM
SUBMITTED BY: Joe Ramos
ANALYZED DATE: 1/9/24

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 11.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW 3	T4003007B	Electrical Conductivity (EC)	EPA 120.1 and	1.27	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	9.41	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	690	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	9.41	mg/L
		Soluble Salts (SALT-SOL)	Calculation	814	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	99.8	97.0 (T4003004A)	96.4 (T4003004A)			0.619
Electrical Conductivity (EC)	99.2			1.54 (T4003002B)	1.52 (Dup)	1.05
Nitrate Nitrogen (NO3-N)	101	97.9 (T4003004B)	95.9 (T4003004B)			2.04
Total Dissolved Solids (TDS)	103			435 (T4003007F)	440 (Dup)	1.14
Total Nitrogen (TN)	104			0.000 (T4003007A)	0.000 (null)	

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ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
TONY L. LOPES DAIRY L.P.
9270 S. Whitworth Road
Gustine, CA 95322

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 1/3/24 9:10 AM
SUBMITTED BY: Joe Ramos
ANALYZED DATE: 1/9/24

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 12.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW 7	T4003007C	Electrical Conductivity (EC)	EPA 120.1 and	1.31	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	15.9	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	640	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	15.9	mg/L
		Soluble Salts (SALT-SOL)	Calculation	841	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	99.8	97.0 (T4003004A)	96.4 (T4003004A)			0.619
Electrical Conductivity (EC)	99.2			1.54 (T4003002B)	1.52 (Dup)	1.05
Nitrate Nitrogen (NO3-N)	101	97.9 (T4003004B)	95.9 (T4003004B)			2.04
Total Dissolved Solids (TDS)	103			435 (T4003007F)	440 (Dup)	1.14
Total Nitrogen (TN)	104			0.000 (T4003007A)	0.000 (null)	

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Compliance Analysis Report

Customer
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Gustine, CA 95322

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 1/3/24 9:10 AM
SUBMITTED BY: Joe Ramos
ANALYZED DATE: 1/9/24

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 11.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW 8	T4003007D	Electrical Conductivity (EC)	EPA 120.1 and	1.80	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	23.6	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	995	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	26.4	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1150	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	99.8	97.0 (T4003004A)	96.4 (T4003004A)			0.619
Electrical Conductivity (EC)	99.2			1.54 (T4003002B)	1.52 (Dup)	1.05
Nitrate Nitrogen (NO3-N)	101	97.9 (T4003004B)	95.9 (T4003004B)			2.04
Total Dissolved Solids (TDS)	103			435 (T4003007F)	440 (Dup)	1.14
Total Nitrogen (TN)	104			0.000 (T4003007A)	0.000 (null)	

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Compliance Analysis Report

Customer
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9270 S. Whitworth Road
Gustine, CA 95322

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 1/3/24 9:10 AM
SUBMITTED BY: Joe Ramos
ANALYZED DATE: 1/9/24

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 11.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW 9	T4003007E	Electrical Conductivity (EC)	EPA 120.1 and	1.30	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	9.46	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	680	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	9.46	mg/L
		Soluble Salts (SALT-SOL)	Calculation	833	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	99.8	97.0 (T4003004A)	96.4 (T4003004A)			0.619
Electrical Conductivity (EC)	99.2			1.54 (T4003002B)	1.52 (Dup)	1.05
Nitrate Nitrogen (NO3-N)	101	97.9 (T4003004B)	95.9 (T4003004B)			2.04
Total Dissolved Solids (TDS)	103			435 (T4003007F)	440 (Dup)	1.14
Total Nitrogen (TN)	104			0.000 (T4003007A)	0.000 (null)	

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Compliance Analysis Report

Customer
TONY L. LOPES DAIRY L.P.
9270 S. Whitworth Road
Gustine, CA 95322

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 1/3/24 9:10 AM
SUBMITTED BY: Joe Ramos
ANALYZED DATE: 1/5/24

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 12.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW 12	T4003007F	Electrical Conductivity (EC)	EPA 120.1 and	0.930	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	4.72	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	435	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	4.72	mg/L
		Soluble Salts (SALT-SOL)	Calculation	595	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	99.8	97.0 (T4003004A)	96.4 (T4003004A)			0.619
Electrical Conductivity (EC)	97.8			0.930 (T4003007F)	0.930 (Dup)	0.065
Nitrate Nitrogen (NO3-N)	101	97.9 (T4003004B)	95.9 (T4003004B)			2.04
Total Dissolved Solids (TDS)	103			435 (T4003007F)	440 (Dup)	1.14
Total Nitrogen (TN)	104			0.000 (T4003007A)	0.000 (null)	

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ELAP Certificate No.: 2714

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Compliance Analysis Report

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9270 S. Whitworth Road
Gustine, CA 95322

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 1/3/24 9:10 AM
SUBMITTED BY: Joe Ramos
ANALYZED DATE: 1/5/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 12.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Calf Barn	T4003007G	Electrical Conductivity (EC)	EPA 120.1 and	1.58	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	14.8	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1010	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	99.8	97.0 (T4003004A)	96.4 (T4003004A)	0.619
Electrical Conductivity (EC)	97.8			0.065
Nitrate Nitrogen (NO3-N)	101	97.9 (T4003004B)	95.9 (T4003004B)	2.04

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2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 1/3/24 9:10 AM
SUBMITTED BY: Joe Ramos
ANALYZED DATE: 1/5/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 12.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
North Domestic	T4003007H	Electrical Conductivity (EC)	EPA 120.1 and	2.64	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	36.7	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1690	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	99.8	97.0 (T4003004A)	96.4 (T4003004A)	0.619
Electrical Conductivity (EC)	97.8			0.065
Nitrate Nitrogen (NO3-N)	101	97.9 (T4003004B)	95.9 (T4003004B)	2.04

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Gustine, CA 95322

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 1/3/24 9:10 AM
SUBMITTED BY: Joe Ramos
ANALYZED DATE: 1/5/24

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 12.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
South Domestic	T4003007I	Electrical Conductivity (EC)	EPA 120.1 and	2.57	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	35.8	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1640	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	99.8	97.0 (T4003004A)	96.4 (T4003004A)	0.619
Electrical Conductivity (EC)	97.8			0.065
Nitrate Nitrogen (NO3-N)	101	97.9 (T4003004B)	95.9 (T4003004B)	2.04

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DENELE ANALYTICAL, INC.
1232 SOUTH AVENUE
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: TONY L. LOPES DAIRY L.P.
Mailing address: 9270 S. Whitworth Road
(not facility address) Gustine CA 95322
Phone#:

Authorized Copy Released To: F&R
Sampled By:
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	1/2	1:00	IW 2		11.8°c						H7
2	1/2	1:10	IW 3		11.0°c						H7
3	1/2	1:15	IW 7		12.0°						H7
4	1/2	1:25	IW 8		11.8°						H7
5	1/2	1:30	IW 9		11.8°						H7
6	1/2	1:40	IW 12		12.0°c						H7
7	1/2	2:00	Calf Barn	DW	12.0°c		X				
8	1/2	2:05	North Domestic	↓	12.2°		X				
9	1/2	2:10	South Domestic	↓	12.0°c		X				
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	*(LABORATORY USE ONLY)* Logged In By:	LABORATORY NUMBER
1st	[illegible]					
2nd	[illegible]	DENELE ANALYTICAL, INC.	1-3-24 0:10			T4003007

Tony L. Lopes 2023 AR Complete

Final Audit Report 2024-04-19

Created:	2024-04-18
By:	Joe Ramos (jramos@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAuqiDy2DSMuBTnOswchRF7r1ybUjCJVcd

"Tony L. Lopes 2023 AR Complete" History

Document created by Joe Ramos (jramos@fragservices.com)
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Signer pandd95@gmail.com entered name at signing as Paul Lopes
2024-04-19 - 6:47:25 AM GMT

Document e-signed by Paul Lopes (pandd95@gmail.com)
Signature Date: 2024-04-19 - 6:47:27 AM GMT - Time Source: server

Agreement completed.
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